Thiriloganathan Manimohan

Manimohan05 | Manimohan | Manimohan | Manimohan | Manimohan |
manimohan517@gmail.com | manimohant.20@uom.lk | +94 771781833

Summary

I'm a highly motivated, confident, and self-directed final-year undergraduate. I have interests in Machine Learning & Deep Learning, Electronics & telecommunications, Robotics & Automation, Web & App Development, UI/UX Designs, and Data science. I'm a flexible person, who is eager to find an internship, where I can gain and enhance my knowledge and skills while being productive and giving my best to the employer. I believe in teamwork and I always consider the best outcome for the team., Take a look at my portfolio.

Education

2021 - present	B.Sc(Hons) in ENTC Engineering. at **University of Moratuwa**	(CGPA : 3.66/4.0)
2020 - 2021	Business Degree in CIMA (Level - 01) at **Wisdom Business Academy**	
2019 - 2020	Human Resource Management at **International Business Management Institute**	
2017 - 2019	G.C.E Advanced Level at **J/Mahajana college**	(3As with Z-score=2.6266)
2005 - 2016	Primary & Secondary at **J/Mahajana college**	(9As)

G.C.E A/L - 3As in Physical science stream and 12 th in district & 116 th in island with Z-score = 2.6266
G.C.E O/L - 9As including ICT and Commerce & Business in English medium.

Image Processing and Machine Vision, Pattern Recognition, Digital Signal Processing, Data Structures and Algorithms(A), Analog Circuit Design, Fundamentals of Computer Organization and Design, Communication Network Engineering (A+), Communication System Engineering(A), Robot Design and Competition, Programming Fundamentals(A-)

Internships & Experiences

- **ML & AI Related research at Singapore University of Technology and Design**
  I am currently doing a six-month internship in Singapore from November 2023 to May 2024. I am working on the following projects
  - Use of advanced vision systems and ML for AM closed loop control
  - Use of ML to identify optimal printing parameters

Projects

- **DENSELY PACKED PRODUCT DETECTION**
  Deep Learning & Image processing and Machine vision | Group | Completed | Duration:- 3 months | Python | This research explores how YOLOv8 object detection can transform retail inventory management by addressing challenges on densely packed supermarket shelves, with a focus on the SKU-110K dataset and broader industry implications.
- **Tamil GPT**
  Machine Learning & Generative AI | Group - Guided | On - Going | Duration:- 6 months | Python | This is a GPT chatbox, which will be able to generate poems with the required feelings using the language handling methods of past writers in the Tamil language.
- **Machine Learning Projects**
  Machine Learning & Deep learning | Induvidual - Guided | Completed| Duration:- 5 months | Python | This includes some machine learning projects I completed while learning about ML concepts| Brain Tumor Detection | Diabetes Detection | Heart Disease Detection | AI Room Booking Chabot | Classification of Arrhythmia Using ECG Data | Human Activity Detection | Medical Chatbot | Wine Quality Prediction

- **Agri Link**
  Webapplication Development | Group - 2023 | Completed| Duration:- 2 months | Salesforce | AgriLink aims to promote fairer trade and a prosperous future for Sri Lanka's agriculture by addressing challenges and promoting direct producer-buyer links.

- **Flower Stock Exchange**
  App Development | Pair - 2023 | Completed| Duration:- 3 months |C++, IntelliJ Clion| This undertaking was completed in the context of the "C++ Workshop" provided by the "London Stock Exchange Group (LSEG)."Developed a trading and exchange application that enables users to engage in buying, selling, and swapping stocks.Integrated sockets to implement virtual client and server systems in the programming.

- **ECG Heart Monitor**
  Bio-Medical | Group | Completed | Duration:- 3 months | Analog project - 2023 | PCB Design | Solidworks | A heart ECG monitor is a device that uses electrodes to capture 3-lead ECG signals. We have applied our analog electronics expertise to enhance signal quality while amplifying the necessary signals. We have to construct an analog front end that accepts data from the ECG probes and outputs a clean waveform of the ECG.

- **LED Flash**
  Electronics | Individual - 2023 | Completed| Duration:- 3 months| PCB Design | Solidworks | This 12V Portable LED Flash initially designed to avoid interruption due to power failures. The key reason to build this Portable LED Flash is to get continuous light during power failures. This is able to act as a mobile power bank and router backup.

- **Network Design and Backbone Design & Network Routing Simulation - OSPF**
  Telecommunication | Group | Completed | Duration:- 3 weeks | Packet Tracer | Backbone network for University of Moratuwa, LAN network for ENTC department and Assign IP addresses and Network Routing Simulation - OSPF

- **Linear Power Supply**
  Electronics | Group | Completed | Duration:- 3 weeks | PCB Design | Input range 16V to 20V ,Output range 8V to 14V and current limiting is 100mA

- **Chat - Application** | **Date: 18th April 2023**
  Web Application - Development | Individual | Completed | Duration:- 1 month | This is a web applications.We can add users and create passwords manually. At the same time, we can send photos.
  To view the app :- Click here
  Test username :- Vadivelu & password :- 1234

- **Autonomus Moving Robot**
  Robotics & Automation | Group | Completed | Duration:- 2 months | Sri Lankan Robotics Challange - 2023 | Line following | wall following | maze solving | Robotics has become increasingly popular, and E-club and the Department of Electronics and Telecommunication Engineering have staged the Sri Lankan Robotics Challenge (SLRC) ("Task") for nine consecutive years.

- **Processor Design & ISA Design**
  Computer Organization & Design | Group | Completed | Duration:- 3 months | RISC - V | Processor design project for Continuous assignment in EN2031 module based on knowledge from Dr. Pasqual's lectures.

- **Maze-solving, line and wall following robot**
  Robotics & Automation | Group | Duration:- 3 months | Completed | Virtual and Physical implementation - 2023 | Line following | wall following | maze solving | Both a simulated and a tangible version of the robot were generated. The simulated iteration was accomplished within Webots, while the physical realization was achieved through the utilization of an Arduino Mega. The robot's design was formulated to cater to both implementations using SolidWorks.(**https://cyberbotics.com**). Webots R2021b version for the simulation.

- **Design of FIR and IIR Filters**
  Signal Processing | Individual - 2022 | Completed | Duration:- 3 months| FIR (Bandpass) | IIR(Bandpass) | The report discusses the procedure used to design FIR and IIR digital filters for prescribed specifications and its implementation on Matlab R2021a. The windowing method is used to truncate the impulse response, and the required digital filter is obtained by applying a bilinear transform to the transfer function of the analog filter.

- **Water Level Detector**
  Telecommunications | Group - 2022 | Completed | Duration:- 3 months | PCB Design | Solidworks | We designed a project using Altium and SolidWorks to detect the water level in tanks to reduce water wastage and power cuts. As part of the course module EN1190 Engineering Design Project, we started the project in the beginning of the semester and completed it at the end of the semester.

Certificates

- **Deep Learning Specialization** - (DeepLearning.AI Stanford University)-Neural Networks and Deep Learning(1/5)
- **Machine Learning Specialization** - (DeepLearning.AI Stanford University)
- **Introduction to FPGA Design for Embedded Systems.** - (University of Colorado Boulder)
- **Introduction to Computer Vision and Image Processing** - (IBM)
- **Foundations of User Experience (UX) Design** - (Google)
- **Salesforce Certified Associate**-(Salesforce Trialhead)
- **System Verilog for ASIC/FPGA Design Simulation**-Department of Electronic Telecommunication Engineering, University of Moratuwa (ENTC, UoM)
- **Front-End Web Development** , **Python Programming** , **Web Design for Beginners** , **Python for beginners** - (Open Learning Platform - University of Moratuwa)
- **Technical Support Fundamentals** - (Google)
- **Salesforce, 23 Badges,9,600 Points,1 Trails** - (Trailblazer)
- **SLRC Advanced Workshop - ROS - 2023** - (University of Moratuwa)
- **KickStarter on Embedded System Design** - (Electronic Club of University of Moratuwa)
- **python for Data science, python Data structure, python core, and SQL** - (Sololearn)
- **python for machinelearning** - (great learning)
- **Data structure and algorithm using C++, Web development, python data Visualization, Python Data Analysis, and Python computervision** - (shapeAI)

Honours, Awards,and Participans

I have some professional memberships, honors,awards, and participants:

- **ICIIS 2023: Special session on advance in 6G Technology** - (Participant)
- **Dean's List- semester-04**
- **Huawei ICT Competition 2022 - 2023** - (Winner)
- **Mahapola Scholarship - (2020)**
- **IEEE Membership-(Professional)** - Membership number - 98709969 - membership certificate
- **National Mathematics Quiz Contest - 2019** - (Finalist)
- **National Science Quiz Contest - 2019** - (First place)
- **3rd place in 30th All island Chemistry Quiz - 2019** - (Institute of Chemistry)
- **Bushman's Thong Awarded Scout(CC and DC)- 2018** - (Honourable Award)
- **Subaha Scholarship - 2018** - (Multi-Talented student)
- **Know Your Rights - 2018** - (Finalist)
- **Tokyo super Quiz Contest - 2016** - (Finalist)
- **National GK Quiz** - (First place in 2010,2012,2013,2014 and 2016)
- **MoraXtreme 7.0 and IEEEXtreme 16.0 - 2022(1288th rank)** - (Participant-TeamM01)
- **Idealize - 2022** - (Finalist)
- **Yarl geek challenge - 2022** - (Semifinalist)
- **Codegoda - 2023** - (Participation)
- **HackX - 2022** - (Participation)

- **RealHack - 2023 March** - (Participants, 22th rank out of 203 teams)
- **Hackforce - 2021** - (Participants)
- **Participant in globalculturaljamboree - (online - 2021)**

Peer Reviews & Publications-Ongoing

Machine learning - Data preprocessing - Publications
We are planning to create a generative AI chatbox that can compose Tamil songs enriched with metaphors, drawing inspiration from the language handling techniques employed by past Tamil writers. There will be 3 publications(Data set, ML model, Generative AI Chatbox) going to be released.

Robotics & Automation - Peer reviews

- Rodney, A. Brooks (1985). "Peer review of Paper: A ROBUST LAYERED CONTROL SYSTEM FOR A MOBILE ROBOT". In: *MASSACHUSETTS INSTITUTE OF TECHNOLOGY ARTIFICIAL INTELLIGENCE LABORATORY Journal* all.full, pp. 1–28. URL: https://github.com/Manimohan05/peer-reviews.git.

Information Technology Skills

Programming Languages	**Python, C/C++,JavaScript/TypeScript,Matlab,LaTeX**: *Advanced* **SQL,Java** : *Intermediate*
Machine Learning Frameworks	**PyTorch, TensorFlow**: *Advanced* **SciKit learn** : *Intermediate*
Frameworks	**React Native, React,Angular**: *Advanced* **Dijango** : *Intermediate*
Data Analysis	**Visual Studio, Git/GitHub, Jupiter Notebook and Colab**: *Advanced* **Wolfram Mathematica** : *Intermediate*
Modeling and Simulation	**Solidworks, Altium and Proteus** : *Advanced* **Quartus, MATLAB, Modelsim, Simulink, Webots, LTSpice and Vivado** : *Advanced*
Designing	**Photoshop,PremierPro** : *Advanced* **AfterEffects** : *Intermediate*
Office Automation	**LaTeX, MS Office (Excel, Word, PowerPoint)** : *Higly Specialized*

Volunteering, Extracurriculars and Other Activities

I am doing volunteer activities, extracurriculars, and other activities:

- **Batch Representative - Department of Electronic and Telecommunications Engineering, University of Moratuwa** - (Semester 04, 05, & 06)
- **Junior & Senior Prefect in J/Mahajana college** - (2014 - 2019)
- **Vedio coordinator in PR branch - 2023/24 Electronic Club - UOM**
- **A Scout - Bushmanthong awarded**-(2010- Ongoing)
- **Member - Hiking club**
- **SLOMF Math Trainers in SLOMF - (from January,2023)**
- **Social media manager in Yarl Salesforce Ohana - (from January,2023)**

- **Demonstrator in SLRC (University of Moratuwa) - (from November,2022)**
- **AIESECer(IGVs) in AIESEC (University of Moratuwa) - (from May,2021)**
- **Teacher in Sasnaka Sansada - (from May,2021)**
- **Volunteer in Mahajanan Volunteers & Sirakukal - (from Nov,2020)**
- **Cricket Player** - (2012 - Ongoing)

References

Dorien Herremans, PhD,
Associate Professor, ISTD & DAI
Director, SUTD Game Lab
Lead, AMAAI (Audio, Music and AI Lab)
Singapore University of Technology and Design
Information Technology and Design Pillar
Email: dorien_herremans@sutd.edu.sg
http://dorienherremans.com

Pablo Valdivia y Alvarado, Ph.D.,
Associate Professor
Engineering Product Development
Director, DManD
Singapore University of Technology and Design
Phone: +65-6499-8883
Email: pablov@sutd.edu.sg
https://dmand.sutd.edu.sg
http://brd.sutd.edu.sg/

Dr. Thayaparan Subramaniam,
B.Sc.Eng (Peradeniya), Ph.D (HKU)
HOD,
Dept. of Electronic and Telecommunication Engineering,
University of Moratuwa,
Katubedda, Moratuwa 10400,
Sri Lanka.
Email: thayaparan@uom.lk

Dr. Uthayasanker Thayasivam,
PhD (U. Georgia), BSc Eng. (Hons) (Moratuwa),
HOD,
Dept. of Computer Science Engineering,
University of Moratuwa,
Katubedda, Moratuwa 10400,
Sri Lanka.
Email: rtuthaya@cse.mrt.ac.lk

Dr. Chamira U. S. Edussooriya
BSc Eng (Moratuwa), MASc (UVic), PhD (UVic), MIEEE
Senior Lecturer,
Dept. of Electronic and Telecommunication Engineering,
University of Moratuwa,
Katubedda, Moratuwa 10400,
Sri Lanka.
Email: chamira@uom.lk

Dr. Ranga Rodrigo,
Senior Lecturer,
Dept. of Electronic and Telecommunication Engineering,
University of Moratuwa,
Katubedda, Moratuwa 10400
Sri Lanka
Email: ranga@uom.lk
Phone: +94 11 264 0422

Dr. Kasun T. Hemachandra,
B.Sc. (Moratuwa), M.Sc. (Alberta) Ph.D. (Alberta)
Senior Lecturer,
Dept. of Electronic and Telecommunication Engineering,
University of Moratuwa,
Katubedda, Moratuwa 10400,
Sri Lanka.
Email: kasunh@uom.lk

Dr. Samiru Gayan,
Senior Lecturer,
Dept. of Electronic and Telecommunication Engineering,
University of Moratuwa,
Katubedda, Moratuwa 10400,
Sri Lanka.
Email: samirug@uom.lk
Phone: +94 77 590 7830

Dr. Sampath K. Perera,
B.Sc. Eng. (Moratuwa), M.E.Sc (Western), P.hD. (RUB), MIEEE,
Senior Lecturer,
Dept. of Electronic and Telecommunication Engineering,
University of Moratuwa,
Katubedda, Moratuwa 10400,
Sri Lanka.
Email: sampathk@uom.lk
Phone: +94 70 572 6264

Dr. Ravi Chaithanya Mysa,
Senior Research Fellow
Singapore University of Technology and Design
8 Somapah Road, Building 1, Room #: 1.706
Singapore 487372
Phone: +65-86579959
Email: ravi_chaithanya@sutd.edu.sg

Last updated: August 3, 2024